

ECON-3210 Applied Econometrics

Review: Matrix and Linear Algebra

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Vector spaces

Matrix algebra

Eigenvalue and Eigenvector

Singular Value Decomposition

Principal Component Analysis

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Vectors and vector space

A **vector** is an ordered list of real numbers. A two-entry column vector $\mathbf{x}$ is

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

The set of all two-entry vectors is called the **vector space** $\mathcal{R}^2$

An n -entry vector (an $n \times 1$ array) of real numbers: $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$

The **real vector space** $\mathcal{R}^n$ consists of all n -element real vectors

In general, a **vector space** $\mathcal{V}$ is the set of **all** column vectors satisfying two properties

- Closure under vector addition: If $\mathbf{x} \in \mathcal{V}$ and $\mathbf{y} \in \mathcal{V}$, then $\mathbf{x} + \mathbf{y} \in \mathcal{V}$
- Closure under scalar multiplication: If $\mathbf{x} \in \mathcal{V}$, then $\alpha\mathbf{x} \in \mathcal{V}$, for any real scalar α

It is easy to check that the space $\mathcal{R}^n$ is indeed a real vector space

Vector operations

Use two-element vectors as example, operations apply to vectors of any dimension. Consider $\mathbf{u}' = [u_1, u_2]$ and $\mathbf{v}' = [v_1, v_2]$

Vector addition: Element-wise addition of two vectors of the same dimension

$$\mathbf{u} + \mathbf{v} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} + \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 \end{bmatrix}$$

Scalar-vector multiplication: $\alpha \mathbf{u} = \alpha \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} \alpha u_1 \\ \alpha u_2 \end{bmatrix}$

Linear combination of vectors: $\alpha \mathbf{u} + \beta \mathbf{v} = \begin{bmatrix} \alpha u_1 + \beta v_1 \\ \alpha u_2 + \beta v_2 \end{bmatrix}$

Example

$$\mathbf{u}' = [2, 5] \text{ and } \mathbf{v}' = [3, 1]$$

$$\mathbf{u} + \mathbf{v} = \begin{bmatrix} 2 + 3 \\ 5 + 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$$

$$0.3\mathbf{u} - 0.5\mathbf{v} = \begin{bmatrix} 0.3 \cdot 2 - 0.5 \cdot 3 \\ 0.3 \cdot 5 - 0.5 \cdot 1 \end{bmatrix} = \begin{bmatrix} -0.9 \\ 1.0 \end{bmatrix}$$

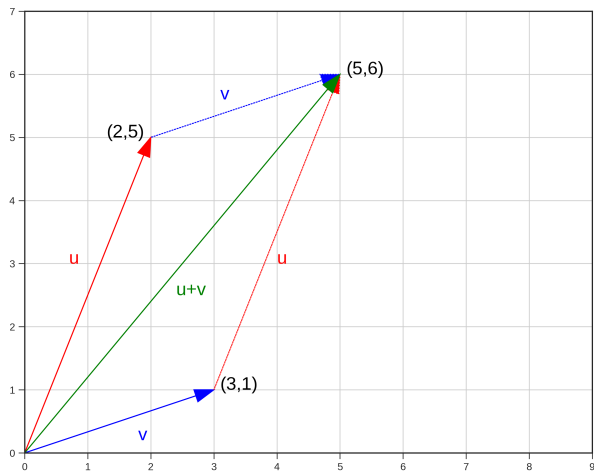


Figure 1: Geometry of vector addition

Inner products and norms

The **inner product** of two vectors, $\mathbf{x}$ and $\mathbf{y}$ in vector space $\mathcal{R}^n$, is defined as

$$\mathbf{x} \cdot \mathbf{y} := x_1y_1 + x_2y_2 + \cdots + x_ny_n = \sum_{i=1}^n x_iy_i$$

Vectors $\mathbf{x}$ and $\mathbf{y}$ are said to be **orthogonal** if their inner product equals zero, i.e., $\mathbf{x} \cdot \mathbf{y} = 0$

Vectors $\mathbf{x}$ and $\mathbf{y}$ are **orthonormal** if $\mathbf{x} \cdot \mathbf{y} = 0$ and $\mathbf{x} \cdot \mathbf{x} = 1$ and $\mathbf{y} \cdot \mathbf{y} = 1$

Example: $\mathbf{x} = [3, -1, 4]'$, $\mathbf{y} = [4, 0, -3]'$ are orthogonal

The Euclidean **norm** of vector $\mathbf{x}$ is given by

$$\|\mathbf{x}\| := \sqrt{\mathbf{x} \cdot \mathbf{x}} = \sqrt{x_1x_1 + x_2x_2 + \cdots + x_nx_n} = \left(\sum_{i=1}^n x_i^2 \right)^{\frac{1}{2}}$$

- The Euclidean norm of a vector is also called the **length** of the vector
- Vector $\mathbf{x}$ is called a **unit vector** if $\|\mathbf{x}\| = 1$
- The Euclidean **distance** of $\mathbf{x}$ and $\mathbf{y}$ is defined as $\|\mathbf{y} - \mathbf{x}\|$

There are other norms, e.g. $\|\mathbf{x}\|_1 = \sum_{i=1}^n |x_i|$ and $\|\mathbf{x}\|_\infty = \max_{i=1, \dots, n} |x_i|$

Example. Two 3-element vectors: $\mathbf{p}' = [3, -1, 4]$, $\mathbf{q}' = [2, 0, -5]$

Inner product: $\mathbf{p} \cdot \mathbf{q} = 3 \cdot 2 + (-1) \cdot 0 + 4 \cdot (-5) = -14$

The length (norm) of $\mathbf{p}$ is $\|\mathbf{p}\| = \sqrt{3 \cdot 3 + (-1) \cdot (-1) + 4 \cdot 4} = \sqrt{26}$

Example. Two 3-element vectors: $\mathbf{u}' = [3, -1, 4]$, $\mathbf{v}' = [4, 0, -3]$ $\mathbf{u}$ and $\mathbf{v}$ are orthogonal, since

$$\mathbf{u} \cdot \mathbf{v} = 3 \cdot 4 + (-1) \cdot 0 + 4 \cdot (-3) = 0.$$

Example. Unit vectors

- $\mathbf{u}' = [\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$ is a unit vector, since $\mathbf{u} \cdot \mathbf{u} = \frac{1}{4} + \frac{1}{4} + \frac{1}{4} + \frac{1}{4} = 1$, so $\|\mathbf{u}\| = \sqrt{1} = 1$
- $\mathbf{e}'_1 = [1, 0, 0, \dots, 0]$, $\mathbf{e}_1$ is called a **standard unit vector**

Linear independence

A **linear combination** of a list of vectors $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\}$ for $k > 0$ in a vector space $\mathcal{V}$ is a vector of the form

$$a_1\mathbf{x}_1 + a_2\mathbf{x}_2 + \dots + a_k\mathbf{x}_k$$

where each $\mathbf{x}_i$ is a vector, each a_i is a real number, and k is an integer

Example: In $\mathcal{R}^3$, $2 \cdot \begin{bmatrix} 7 \\ -3 \\ 10 \end{bmatrix} + 4 \cdot \begin{bmatrix} 1 \\ 4 \\ 2 \end{bmatrix} + 10 \cdot \begin{bmatrix} 5 \\ 1 \\ 0 \end{bmatrix} - 0.5 \cdot \begin{bmatrix} 7 \\ -1 \\ 30 \end{bmatrix}$

By definition, a linear combination of vectors is also in vector space $\mathcal{V}$

A set of vectors $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\}$ is called **linearly independent** if and only if $a_1\mathbf{x}_1 + a_2\mathbf{x}_2 + \dots + a_k\mathbf{x}_k = \mathbf{0}$ implies $a_1 = a_2 = \dots = a_k = 0$

Is the set of vectors in previous example linearly independent?

A set of vectors $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\}$ is **linearly dependent** if, for some real scalars $a_1, a_2, \dots, a_k$, not all are zero, $a_1\mathbf{x}_1 + a_2\mathbf{x}_2 + \dots + a_k\mathbf{x}_k = \mathbf{0}$

A set of vectors is linearly dependent if one vector equals the linear combination of other vectors

Example: vectors $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}$, $\mathbf{v}_3 = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$ are linearly independent.

(Check it using the definition)

Let $S = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\}$ be a subset of a vector space $\mathcal{V}$, let **span**(S) be the set of all linear combinations of vectors in S ,

$$\text{span}(S) = \{a_1\mathbf{x}_1 + a_2\mathbf{x}_2 + \dots + a_k\mathbf{x}_k \mid \forall a_i \in \mathcal{R}, i = 1, \dots, k\}.$$

S is said to **span** $\mathcal{V}$ if $\text{span}(S) = \mathcal{V}$, that is,

- S spans $\mathcal{V}$ if each vector in $\mathcal{V}$ is a linear combination of vectors in S

Basis

A list of vectors, $S = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\}$ in vector space $\mathcal{V}$, is called a **basis** for $\mathcal{V}$ if

- $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\}$ are linearly independent, and
- S spans $\mathcal{V}$, that is, $\text{span}(S) = \mathcal{V}$

Any vector of $\mathcal{V}$ is a linear combination of vectors of basis S in one and only one way

Proof. Suppose statement is not true, then any vector $\mathbf{v}$ can be written as $\mathbf{v} = a_1\mathbf{x}_1 + a_2\mathbf{x}_2 + \dots + a_n\mathbf{x}_n$ and $\mathbf{v} = b_1\mathbf{x}_1 + b_2\mathbf{x}_2 + \dots + b_n\mathbf{x}_n$

This means $\mathbf{0} = (a_1 - b_1)\mathbf{x}_1 + (a_2 - b_2)\mathbf{x}_2 + \dots + (a_n - b_n)\mathbf{x}_n$, which is only true if $a_1 = b_1, a_2 = b_2, \dots, a_n = b_n$, since vectors $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n$ are linearly independent □

Put it differently, the list of vectors $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\}$ is a basis for $\mathcal{V}$ if, for any $\mathbf{v} \in \mathcal{V}$,

$$a_1\mathbf{x}_1 + a_2\mathbf{x}_2 + \dots + a_k\mathbf{x}_k = \mathbf{v}$$

has unique solution of values of $\{a_1, \dots, a_k\}$

Example: The **standard basis** for vector space $\mathcal{R}^n$. The set of unit vectors $\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$ in $\mathcal{R}^n$ forms a basis for vector space $\mathcal{R}^n$, where the i -th entry of each n -entry vector $\mathbf{e}_i$ is one and the rest elements are zeros

That is, the standard basis for vector space $\mathcal{R}^n$ is:

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \mathbf{e}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \dots, \mathbf{e}_n = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}.$$

Example: In $\mathcal{R}^3$, vector $\mathbf{v} = \begin{bmatrix} 7 \\ -3 \\ 10 \end{bmatrix}$ is the linear combination of vectors of the standard basis:

$$\mathbf{v} = 7 \cdot \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} - 3 \cdot \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + 10 \cdot \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

Standard unit vectors are linearly independent

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \mathbf{e}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

It is straightforward to check that $a_1\mathbf{e}_1 + a_2\mathbf{e}_2 + a_3\mathbf{e}_3 = \mathbf{0}$ only if $a_1 = a_2 = a_3 = 0$

This verifies that the set of standard unit vectors indeed constitutes a basis

Matrix algebra

Matrices

An $m \times n$ **matrix** $\mathbf{A}$ consists of m rows and n columns of (real) numbers

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}.$$

If $m = n$, $\mathbf{A}$ is called a square matrix

The **diagonal** of a matrix $\mathbf{A}$ is the vector consisting of a_{ii}

The **trace** of square matrix $\mathbf{A} = [a_{ij}]_{n \times n}$ is $\sum_{i=1}^n a_{ii}$

The **identity matrix** $\mathbf{I}$ is $n \times n$ with ones on the diagonal and zeros off diagonal

The **transpose** of matrix $\mathbf{A}$ is $\mathbf{A}^T$, the i -th row of $\mathbf{A}$ is the i -th column of $\mathbf{A}^T$

Example. $\mathbf{A} = \begin{bmatrix} 1 & 4 \\ 0 & -3 \\ 1 & 1 \end{bmatrix}$, then $\mathbf{A}^T = \begin{bmatrix} 1 & 0 & 1 \\ 4 & -3 & 1 \end{bmatrix}$

The **rank** of a matrix is the length of a longest linearly independent list of columns

Matrix operations

Matrix addition: $\mathbf{A} + \mathbf{B} = [a_{ij} + b_{ij}]_{m \times n}$

Matrix multiplication: Let $\mathbf{A}$ be $m \times n$ and $\mathbf{B}$ be $n \times p$. The multiplication of the matrices is $\mathbf{C} = \mathbf{AB}$, where $c_{ij} = \sum_{k=1}^n a_{ik}b_{kj}$ Matrix $\mathbf{C}$ is $m \times p$

For multiplication, number of columns of the first matrix must equal number of rows of the second matrix

Example:

$$\mathbf{AA}^T = \begin{bmatrix} 1 & 4 \\ 0 & -3 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 4 & -3 & 1 \end{bmatrix} = \begin{bmatrix} 17 & -12 & 5 \\ -12 & 9 & -3 \\ 5 & -3 & 2 \end{bmatrix}$$

Properties of matrix multiplication:

- $\mathbf{AB} \neq \mathbf{BA}$
- $\mathbf{A}(\mathbf{B} + \mathbf{C}) = \mathbf{AB} + \mathbf{AC}$
- $(\mathbf{B} + \mathbf{C})\mathbf{A} = \mathbf{BA} + \mathbf{CA}$
- $\mathbf{A}(\mathbf{BC}) = (\mathbf{AB})\mathbf{C}$
- $\mathbf{IA} = \mathbf{AI} = \mathbf{A}$
- $(\mathbf{AB})^T = \mathbf{B}^T \mathbf{A}^T$

The **power** of matrix is $\mathbf{A}^n = \underbrace{\mathbf{AA} \cdots \mathbf{A}}_{n \text{ times}}$

It can be shown that $\mathbf{A}^m \mathbf{A}^n = \mathbf{A}^{m+n}$ and $(\mathbf{A}^m)^n = \mathbf{A}^{mn}$

Matrix $\mathbf{A}$ is said to be **orthogonal** if $\mathbf{A}^T \mathbf{A} = \mathbf{I}$

Matrix inverse

The **inverse matrix** of an $n \times n$ square matrix $\mathbf{A}$ is $\mathbf{A}^{-1}$ that satisfies the condition $\mathbf{A}^{-1}\mathbf{A} = \mathbf{A}\mathbf{A}^{-1} = \mathbf{I}$

If matrix $\mathbf{A}$ has no inverse matrix, then $\mathbf{A}$ is called a **singular matrix**

Example: $\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, then $\mathbf{A}^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$, which exists iff $(ad - bc) \neq 0$

Orthogonal matrix: If $\mathbf{A}$ is orthogonal, then $\mathbf{A}^T = \mathbf{A}^{-1}$

Properties of matrix inverse

1. $(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$
2. $(\mathbf{A}^{-1})^{-1} = \mathbf{A}$
3. $(\mathbf{A}^{-1})^T = (\mathbf{A}^T)^{-1}$

Proof.

1. Let $\mathbf{C} = (\mathbf{AB})^{-1}$, then $\mathbf{CAB} = \mathbf{I}$. Postmultiply both sides by $\mathbf{B}^{-1}\mathbf{A}^{-1}$, then $\mathbf{CABB}^{-1}\mathbf{A}^{-1} = \mathbf{IB}^{-1}\mathbf{A}^{-1}$, result follows from properties of matrix multiplication
2. By definition, $(\mathbf{A}^{-1})^{-1}\mathbf{A}^{-1} = \mathbf{I}$, postmultiply $\mathbf{A}$ in both sides, it becomes $(\mathbf{A}^{-1})^{-1}\mathbf{A}^{-1}\mathbf{A} = \mathbf{IA}$. Result follows from properties of matrix multiplication
3. By definition, $(\mathbf{A}^{-1}\mathbf{A})^T = \mathbf{I}^T = \mathbf{I}$. From the properties of matrix multiplication, $(\mathbf{A}^{-1}\mathbf{A})^T = \mathbf{A}^T(\mathbf{A}^{-1})^T = \mathbf{I}$. By the definition of inverse, $(\mathbf{A}^{-1})^T$ is the inverse of $\mathbf{A}^T$, therefore $(\mathbf{A}^T)^{-1} = (\mathbf{A}^{-1})^T$

Determinants

The **determinant** of a square matrix is the "volume" of a box in n -dimensional vector space. Only square matrices have determinants

We use $|\mathbf{A}|$ to denote the determinant of matrix $\mathbf{A}$

Example: Determinant of matrix $\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ is $\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21}$

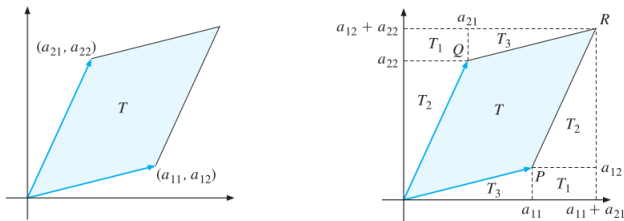


Figure 2: Area T equals the absolute value of determinant $a_{11}a_{22} - a_{12}a_{21}$

A 3×3 matrix $\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$. Its determinant is

$$\begin{aligned} |\mathbf{A}| &= \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = \\ &(-1)^{1+1}a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} + (-1)^{1+2}a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + (-1)^{1+3}a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} \\ &= a_{11}a_{22}a_{33} - a_{11}a_{23}a_{32} + a_{12}a_{23}a_{31} - a_{12}a_{21}a_{33} + a_{13}a_{21}a_{32} - a_{13}a_{22}a_{31} \end{aligned}$$

Determinant $|M_{11}| = \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}$ is called the **minor** of element a_{11} . It is obtained by deleting the 1st row and the 1st column of $|\mathbf{A}|$

Similarly, $|M_{12}| = \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix}$ is the minor of a_{12} . $|M_{13}| = \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$ is the minor of a_{13}

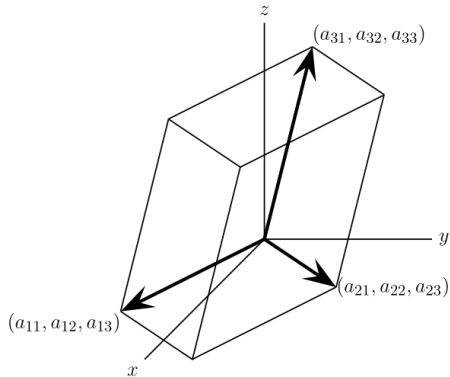


Figure 3: The box formed from the rows of $\mathbf{A}$: volume = absolute value of determinant

For any matrix $\mathbf{A} = [a_{ij}]_{n \times n}$, let $|M_{ij}|$ be the minor of element a_{ij}

Determinant of an $n \times n$ matrix $\mathbf{A}$

The determinant of matrix $\mathbf{A}$ can be found by the Laplace expansion of any row or any column

Expansion by any i th row: $|\mathbf{A}| = \sum_{j=1}^n a_{ij}(-1)^{i+j}|M_{ij}|$

Expansion by any j th column: $|\mathbf{A}| = \sum_{i=1}^n a_{ij}(-1)^{i+j}|M_{ij}|$

Define **cofactor** of element a_{ij} as $|C_{ij}| := (-1)^{i+j}|M_{ij}|$. Then $|\mathbf{A}| = \sum_{j=1}^n a_{ij}|C_{ij}|$

Properties of determinants

- $|\mathbf{A}| = |\mathbf{A}^T|$
- Interchanging any two rows (or any two columns) of matrix alters the sign, but not the numerical value, of determinant of the matrix
- Multiplying any one row (or any one column) by a scalar k changes determinant value k -fold
- Adding (subtracting) a multiple of any row to (from) another row leaves the value of determinant unaltered. The same holds true if we replace the word row by column
 - E.g.,
$$\begin{vmatrix} a & b \\ c + ka & d + kb \end{vmatrix} = \begin{vmatrix} a & b \\ c & d \end{vmatrix}$$
- If one row (or column) is a multiple of another row (or column), the value of the determinant will be zero
- $|\mathbf{AB}| = |\mathbf{A}| \cdot |\mathbf{B}|$ and $|\mathbf{A} + \mathbf{B}| \neq |\mathbf{A}| + |\mathbf{B}|$

Existence of matrix inverse

For an $n \times n$ matrix **A**, the following statements are equivalent. That is, the following statements are either all true or all false

- **A** is invertible (nonsingular)
- The rank of **A** is n
- The columns of **A** are linearly independent
- The determinant of **A** is not zero

Find matrix inverse

Matrix inverse can be computed with determinants, based on the **Cramer's rule** to solve the linear system $\mathbf{Ax} = \mathbf{b}$

Cramer's rule for solving system linear equations $\mathbf{Ax} = \mathbf{b}$:

$$x_i = \frac{\det(\mathbf{A}_i)}{\det(\mathbf{A})}.$$

$\mathbf{A}_i$ is a matrix, its i -th column is $\mathbf{b}$, and all other columns are the same as $\mathbf{A}$

The ij -th element of $\mathbf{A}^{-1}$ is the j -th solution to $\mathbf{Ax} = \mathbf{e}_j$,

- where $\mathbf{e}_j$ is the j -th unit vector: elements are zero except that j -th element is one

Can also Gauss-Jordan method, see any linear algebra textbook

Eigenvalue and Eigenvector

Motivating example

Motivating example: maximizing a symmetric quadratic form. Let $\mathbf{A}$ be symmetric matrix and $\mathbf{x}$ a column vector, the problem is

$$\max_{\mathbf{x}} \mathbf{x}^T \mathbf{A} \mathbf{x} \quad \text{subject to} \quad \mathbf{x}^T \mathbf{x} = 1$$

Let λ be the Lagrange multiplier, the necessary conditions for an extremum are

$$2(\mathbf{A}\mathbf{x} - \lambda\mathbf{x}) = \mathbf{0}$$

λ is an eigenvalue of matrix $\mathbf{A}$

Eigenvalues and eigenvectors

Scalars λ and non-zero vectors $\mathbf{x}$ are respectively called **eigenvalues** and **eigenvectors** of an $n \times n$ matrix $\mathbf{A}$ if they satisfy

$$\mathbf{Ax} = \lambda\mathbf{x}.$$

Eigen is a German word, it means *owned by* or *peculiar to*

The set of distinct eigenvalues of matrix $\mathbf{A}$, denoted by $\sigma(\mathbf{A})$, is called the **spectrum** of $\mathbf{A}$

$$\lambda \in \sigma(\mathbf{A}) \iff (\mathbf{A} - \lambda\mathbf{I})\mathbf{x} = \mathbf{0} \iff \mathbf{A} - \lambda\mathbf{I} \text{ is singular} \iff \det(\mathbf{A} - \lambda\mathbf{I}) = 0$$

Note, $\mathbf{Ax} - \lambda\mathbf{x} = \mathbf{0}$ implies $(\mathbf{A} - \lambda\mathbf{I})\mathbf{x} = \mathbf{0}$

Example. Find eigenvalues of matrix $\mathbf{A} = \begin{bmatrix} 7 & -2 \\ 4 & 1 \end{bmatrix}$

Eigenvalues satisfy $\det(\mathbf{A} - \lambda \mathbf{I}) = 0$

$$\text{Determinant } \begin{vmatrix} 7 - \lambda & -2 \\ 4 & 1 - \lambda \end{vmatrix} = (7 - \lambda)(1 - \lambda) + 6 = (\lambda - 3)(\lambda - 5) = 0$$

Eigenvalues are $\lambda = 3$ and $\lambda = 5$

Let $p(\lambda) = (\lambda - 3)(\lambda - 5)$, it is called the **characteristic polynomial** for $\mathbf{A}$

In general, the eigenvalues of matrix $\mathbf{A}$ are the solution to the **characteristic equation** $p(\lambda) = 0$

Exercise. Find the eigenvalues of $\mathbf{A} = \begin{bmatrix} 7 & -4 \\ 5 & -2 \end{bmatrix}$

Some facts of eigenvalues

1. **Eigenvalue, trace, determinant.** Let $\lambda_1, \lambda_2, \dots, \lambda_n$ be eigenvalues of an $n \times n$ matrix $\mathbf{A}$. Then
 - $\text{trace } \mathbf{A} = \lambda_1 + \lambda_2 + \dots + \lambda_n$
 - $\det(\mathbf{A}) = \lambda_1 \cdot \lambda_2 \cdot \dots \cdot \lambda_n$
2. Eigenvalues of a (either upper or lower) triangular matrix are the diagonal elements $a_{11}, a_{22}, \dots, a_{nn}$
3. Eigenvalues of a diagonal matrix are its diagonal elements

Singular Value Decomposition

Singular value and singular vector

For an $m \times n$ matrix $\mathbf{A}$, let $\lambda_1, \lambda_2, \dots, \lambda_n$ be the eigenvalues of matrix $\mathbf{A}^T \mathbf{A}$. Then, $\sigma_1 = \sqrt{\lambda_1}, \sigma_2 = \sqrt{\lambda_2}, \dots, \sigma_n = \sqrt{\lambda_n}$ are called the **singular values** of matrix $\mathbf{A}$

Singular values are always non-negative

An orthonormal eigenvector ($\mathbf{v}_j$) of $\mathbf{A}^T \mathbf{A}$ is called a **right-hand singular vectors** of $\mathbf{A}$

An orthonormal eigenvector ($\mathbf{u}_j$) of $\mathbf{A} \mathbf{A}^T$ is called a **left-hand singular vectors** of $\mathbf{A}$

It can be show: $\mathbf{A} \mathbf{v}_j = \sigma_j \mathbf{u}_j$

Singular value decomposition

For each $m \times n$ real matrix $\mathbf{A}$ of rank r , there are orthogonal matrices $\mathbf{U}_{m \times m}$ and $\mathbf{V}_{n \times n}$ and there is a diagonal matrix $\mathbf{S}_{r \times r} = \text{diag}(\sigma_1, \sigma_2, \dots, \sigma_r)$ with $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0$, such that

$$\mathbf{A} = \mathbf{U} \begin{bmatrix} \mathbf{S} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix}_{m \times n} \mathbf{V}^T.$$

- The equation above is called the **singular value decomposition (SVD)** of $\mathbf{A}$
- $\sigma_1, \dots, \sigma_r$ are **singular values** of $\mathbf{A}$
- Columns of $\mathbf{U}$ are the left-hand singular vectors of $\mathbf{A}$
- Columns of $\mathbf{V}$ are the right-hand singular vectors of $\mathbf{A}$

Let $\mathbf{u}_i$ be the i -th vector of $\mathbf{U}$, and $\mathbf{v}_j$ the j -th vector of $\mathbf{V}$

Pieces of the SVD:

$$\mathbf{A} = \sigma_1 \mathbf{u}_1 \mathbf{v}_1^T + \sigma_2 \mathbf{u}_2 \mathbf{v}_2^T + \cdots + \sigma_r \mathbf{u}_r \mathbf{v}_r^T.$$

- Each outer product $\mathbf{u}_j \mathbf{v}_j^T$ is an $m \times n$ matrix
- The equation above reduces a complex matrix to simpler ones (that have low ranks)
- Matrix $\mathbf{A}$ can be approximated by the sum of the first several terms on the right-hand side, because σ_1 is the largest hence the first term $\sigma_1 \mathbf{u}_1 \mathbf{v}_1^T$ is the most important

The SVD of any matrix always exists and is unique

An example

$$\mathbf{A} = \begin{bmatrix} 3 & 0 \\ 4 & 5 \end{bmatrix}. \text{ Its rank is 2. Then, } \mathbf{A}^T \mathbf{A} = \begin{bmatrix} 25 & 20 \\ 20 & 25 \end{bmatrix}$$

$\mathbf{A}^T \mathbf{A}$ has two eigenvalues: $\lambda_1 = 45$, $\lambda_2 = 5$ (by the method shown earlier). Thus, $\sigma_1 = \sqrt{45}$, $\sigma_2 = \sqrt{5}$

The eigenvectors of $\mathbf{A}^T \mathbf{A}$ are $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} -1 \\ 1 \end{bmatrix}$. Then, The right orthonormal eigenvectors are $\mathbf{v}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $\mathbf{v}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 1 \end{bmatrix}$, which are the right-hand singular vectors

The left-hand singular vectors are $\mathbf{u}_1 = \frac{\mathbf{A}\mathbf{v}_1}{\sigma_1}$, $\mathbf{u}_2 = \frac{\mathbf{A}\mathbf{v}_2}{\sigma_2}$

The SVD of **A** is **A** = **USV**^T, with

$$\mathbf{U} = \frac{1}{\sqrt{10}} \begin{bmatrix} 1 & -3 \\ 3 & 1 \end{bmatrix}, \mathbf{S} = \begin{bmatrix} \sqrt{45} & 0 \\ 0 & \sqrt{5} \end{bmatrix}, \text{ and } \mathbf{V} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$$

For large matrices, the SVD is calculated in alternative more efficient ways

Principal Component Analysis

Principal Component Analysis

Principal Component Analysis (PCA) refers to the process of finding the principal components of a data matrix, and the subsequent uses of principal components

Principal components of a data matrix are low-dimensional representations of the data matrix that contains as much as possible of the variation of the data matrix

It is an application of SVD

We illustrate the method from the matrix algebra point of view

A data matrix $\mathbf{X}$ has n rows and m columns

- A column represents a variable (a feature), e.g., income, age, education
- A row represents a data observation, e.g., a person's income, age, education

The **mean** of the k -th column: $\bar{x}_k = \frac{1}{n}(x_{1k}, x_{2k}, \dots, x_{nk})$

Let column vector $\mathbf{a}_k = \mathbf{x}_k - \bar{x}_k$. It has n elements

Let $\mathbf{A} = [\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_m]$. It's an $n \times m$ matrix

That is, matrix $\mathbf{A}$ is the demeaned matrix $\mathbf{X}$

- The mean value of each column of $\mathbf{A}$ equals zero

The covariance matrix of data $\mathbf{X}$ is

$$\text{Var}(\mathbf{X}) = \frac{1}{n-1} \mathbf{A}^T \mathbf{A},$$

or written in a long form,

$$[\text{Var}(\mathbf{X})]_{ij} = \frac{1}{n-1} (a_{1i} \cdot a_{1j} + a_{2i} \cdot a_{2j} + \cdots + a_{ni} \cdot a_{nj}),$$

for $i = 1, \cdots, m$ and $j = 1, \cdots, m$

The k -th value on the diagonal of covariance matrix is the variances of $\mathbf{x}_k$

We want to find principal components that contain the variation of data matrix as much as possible

The first principal component

The first principal component of a data matrix $\mathbf{A} = [\mathbf{a}_1 \mathbf{a}_2 \cdots \mathbf{a}_m]$ is the normalized linear combination of the features

$$\mathbf{z}_1 = u_{11}\mathbf{a}_1 + u_{21}\mathbf{a}_2 + \cdots + u_{m1}\mathbf{a}_m,$$

where coefficients are normalized as $(u_{11}^2 + u_{21}^2 + \cdots + u_{m1}^2) = 1$

- $\mathbf{z}_1$ is a column vector with n elements

The i -th element is $z_{i1} = u_{11}a_{i1} + u_{21}a_{i2} + \cdots + u_{m1}a_{im}$

We look for the linear combination that has the largest variance by solving:

$$\max_{\{\mathbf{u}\}} \left[\frac{1}{n-1} \sum_{i=1}^n \left(\sum_{j=1}^m u_{j1} a_{ij} \right)^2 \right] \text{ subject to } \sum_{j=1}^m u_{j1}^2 = 1.$$

In the above problem, the objective function is the variance of $\mathbf{z}_1$

The second principal component and more

The second principal component is also a linear combination of m features:

$$\mathbf{z}_2 = u_{12}\mathbf{a}_1 + u_{22}\mathbf{a}_2 + \cdots + u_{m2}\mathbf{a}_m$$

Coefficients can be also obtained as those in the first principal component

The additional constraint is:

- vector $\mathbf{u}_1 = [u_{11}, u_{21}, \cdots u_{m1}]^T$ is orthogonal to $\mathbf{u}_2 = [u_{12}, u_{22}, \cdots u_{m2}]^T$

In total, there are at most $\min(n - 1, m)$ principal components

It turns out:

- vectors $\mathbf{u}_1$ and $\mathbf{u}_2$ are left-hand singular vectors associated with the larger singular values of matrix $\mathbf{A}$
- variances of principal components are the squares of the singular values of $\mathbf{A}$

Further reading

- [Introduction to Numpy](#)
- Quantitative economics with Python, Section 4: [linear algebra](#)
- [Introduction to Applied Linear Algebra](#), Boyd and Vandenberghe
- Linear Algebra and Its Applications, 4th edition, Gilbert Strang